UMKC Impact & Translation Series

Becoming an Academic Entrepreneur

Understanding one pathway for moving your research into real-world use

Research at UMKC is grounded in discovery. But for many faculty, the deeper motivation behind that work is impact - improving lives, influencing practice, and solving real-world problems beyond the university.

In many cases, that kind of impact requires translation. Academic entrepreneurship is one pathway for doing that. It represents a structured way to move research into real-world use by building something new around it.

At UMKC, we're committed to fostering a supportive environment for researchers who wish to translate their scientific discoveries into real-world applications. This guide will walk you through what it means to be an academic entrepreneur, the benefits, challenges, and how UMKC can support you on this journey.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: what does it mean to be an "academic entrepreneur."

What is Academic Entrepreneurship?

Academic entrepreneurship is one of several pathways that can move research into real-world use. Others include licensing, partnerships, and adoption in practice.

Academic entrepreneurship involves the commercialization of innovations developed within academic settings. It's about taking your research beyond the lab and into the marketplace, where it can make a tangible impact on society. This can involve patenting your discoveries, forming startups, or partnering with existing companies to develop and market new technologies.

In today's fast-paced, technology-driven world, the traditional boundaries of academia are expanding. The internet has democratized knowledge, making it accessible to a global audience.

In some cases, translating research into use allows faculty to more directly contribute to public understanding, practice, and application.

How Academic Entrepreneurship Can Increase Your Work's Impact

The decision to pursue entrepreneurship as an academic can be highly rewarding. Commercializing research allows faculty to maximize the impact of their discoveries, ensuring that valuable

knowledge does not remain confined to academic journals but is instead used to solve pressing problems.

Examples are unique to each individual and setting, but can include:

- **Increased Impact** - Your research can reach a wider audience and have a more significant impact on society.
- **Professional Fulfillment** - Engaging in entrepreneurship can offer personal satisfaction and a sense of accomplishment as you see your ideas come to life.
- **Support for Tenure** – While not explicitly stated in most P&T rules, efforts that result in successful patents, revenue, and impact can provide excellent support for a tenure review.
- **Career Opportunities** - Innovation opens up new career paths within and outside academia, providing diverse professional experiences.
- **Economic Benefits** - Successful commercialization can generate revenue and attract funding for further research.

Challenges and How to Overcome Them

Like any pathway to impact, academic entrepreneurship comes with tradeoffs, constraints, and decisions that shape how and whether it makes sense for your work. Becoming an academic entrepreneur presents unique challenges that differ from those faced by traditional startup founders.

While faculty members have access to cutting-edge research and institutional resources, they must also navigate intellectual property policies, secure funding, and balance commercialization efforts with their academic responsibilities.

Furthermore, venture investors often expect full-time commitment, creating a difficult decision for those wanting to remain in academia. Understanding these challenges in advance and developing strategies to overcome them can significantly improve the chances of a successful transition from research to commercialization. The following sections address key challenges and provide actionable solutions to help faculty entrepreneurs navigate this complex landscape.

Intellectual Property & University Policies

Navigating the complexities of intellectual property (IP) ownership is one of the first challenges faculty members face when considering commercialization. Universities often retain ownership of innovations developed using institutional resources. However, researchers can negotiate licensing rights, allowing them to commercialize their work while still adhering to university policies.

Balancing academic responsibilities with entrepreneurship requires careful planning. Faculty members must ensure that their startup activities align with university conflict-of-interest policies while managing their time effectively. Transparency is key, as institutions have policies in place to guide faculty who are actively engaged in commercial ventures.

Protecting research through patents or other legal mechanisms before public disclosure is essential. Completing an Invention Disclosure Form (available from UMKC's Office of Technology Transfer) is the first step to protecting your innovation and aligns UMKC resources to manage IP concerns, understand commercialization pathways, and develop a protection strategy for IP.

Funding & Grants

Securing funding is a critical step in turning research into a viable commercial product. Many academic entrepreneurs rely on a combination of grants, investments, and institutional resources to support their ventures.

Federal programs such as Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) grants provide funding specifically for early-stage commercialization efforts.

Additionally, UMKC offers internal funding opportunities to support innovation through programs at the UMKC Innovation Center including Digital Sandbox and through the Entrepreneurship Innovation Grant (EIG) and Regnier Venture Creation Challenge administered by the Bloch School.

It is important to note that traditional research grants are typically restricted to academic purposes and cannot always be used for commercialization. Faculty must explore specific translational research grants or industry partnerships that allow for commercialization-related activities.

When considering equity funding for an invention, one significant challenge is that most venture investors expect founders to be full-time entrepreneurs, which can be difficult for faculty members who wish to retain their academic roles. To address this, faculty can explore models where they serve as scientific advisors while bringing in an operational co-founder to manage the company. Alternatively, some universities offer policies that allow faculty members to take a leave of absence or reduce teaching loads to dedicate more time to their startup while maintaining their university affiliation.

The UMKC Office of Commercialization provides guidance and connections to funding sources that support technology transfer and startup development.

Time Management for Faculty Entrepreneurs

Balancing entrepreneurship with academic responsibilities can be one of the biggest challenges for faculty members. Teaching, research, and administrative duties require significant time investment, making it difficult to focus on commercialization efforts. Faculty members interested in entrepreneurship should consider strategies such as reducing teaching loads, delegating administrative responsibilities, and setting clear boundaries between academic and startup commitments.

Time management strategies include prioritizing tasks, leveraging university resources such as business development support, and collaborating with co-founders or experienced entrepreneurs to share responsibilities.

Many universities, including UMKC, offer policies that allow faculty members to take a leave of absence or transition into part-time roles while developing their startups. Understanding these options early can help faculty members create a sustainable balance between academic and business pursuits.

Where Academic Entrepreneurship Fits Among Translation Pathways

There are multiple avenues for faculty members to bring their research innovations to market. Starting a company is one pathway; it is not always the right one

Many researchers successfully commercialize their discoveries by licensing their technology to established companies, forming industry partnerships, or participating in joint ventures. Each pathway has its own set of advantages and considerations, from retaining control over the innovation to leveraging existing market expertise and infrastructure. Faculty members must evaluate these options carefully based on their goals, resources, and long-term commitment to commercialization.

The following sections explore various commercialization strategies to help faculty determine the best approach for translating their research into real-world impact.

Faculty-Industry Collaboration Models

Collaborating with industry partners can accelerate the commercialization of academic research while providing valuable insights into real-world applications. Faculty members can engage with companies through sponsored research agreements, licensing deals, joint ventures, or consulting roles. These collaborations help bridge the gap between academia and the private sector, ensuring that innovations are developed in a way that meets market needs.

Industry collaborations can also provide funding for research that may not qualify for traditional academic grants. Companies often look for university partnerships to leverage cutting-edge research while providing financial resources and commercial expertise. Establishing long-term relationships with industry stakeholders can lead to continued support for faculty-led innovations and facilitate knowledge exchange that benefits both academia and business.

Business Formation & Strategy

One of the key decisions in the commercialization process is whether to license technology to an existing company or form a startup. Licensing allows faculty members to leverage an established company's resources for product development and distribution, reducing the burden of building a business from scratch. However, starting a company offers greater control over the innovation process and potential long-term financial benefits.

For those choosing to launch a startup, selecting the right legal structure is crucial. The decision between forming a Limited Liability Company (LLC), a C-Corporation, or another entity depends on funding goals, ownership structure, and operational needs. Working with legal and business experts can help faculty entrepreneurs make informed decisions about business formation.

Building a strong team is equally important. Faculty members may need to collaborate with co-founders, business professionals, or industry experts to complement their technical expertise. Many successful academic startups involve partnerships between researchers and experienced entrepreneurs who can handle the business and operational aspects of the company.

UMKC's Office of Commercialization works with researchers to identify and connect with entrepreneurs who can take their research public. This allows faculty members to maintain their academic roles while serving as scientific advisors to the company. By leveraging this approach, faculty can focus on advancing research while an experienced entrepreneur leads commercialization efforts.

Industry & Market Considerations

Understanding market demand and industry trends is essential for successful commercialization. Many academic entrepreneurs face the challenge of working in isolation from potential customers, which can lead to misalignment between research innovations and market needs. Engaging with industry partners early in the process can provide insights into potential applications, regulatory requirements, and commercialization pathways.

Conducting market research is a crucial step in validating the commercial viability of an innovation. This may involve customer interviews, competitive analysis, and pilot studies. Universities, including UMKC, offer resources such as business development workshops and mentorship programs to help faculty assess market potential and refine commercialization strategies.

Regulatory & Compliance

Regulatory considerations vary by industry and can significantly impact the commercialization timeline. Faculty members in fields such as healthcare, biotechnology, and engineering must navigate compliance requirements, including FDA approvals, HIPAA regulations, and environmental safety standards. Understanding these requirements early on can prevent costly delays in product development.

Additionally, tax and legal considerations should not be overlooked. Faculty members who receive startup revenue or hold equity in a company should consult with financial and legal professionals to ensure compliance with tax regulations.

Engaging Graduate Students in Entrepreneurship

Graduate students are often at the forefront of academic research, conducting experiments, analyzing data, and identifying novel findings that may have commercial potential. Their proximity

to the research process makes them ideal candidates for spotting new ideas that could lead to startup opportunities.

Faculty members can leverage graduate students as a source of innovation by fostering a culture that encourages entrepreneurship within the lab. This can be done by introducing students to concepts of commercialization, market validation, and technology transfer early in their research journey. Encouraging students to think beyond academic publications and consider the practical applications of their work can lead to groundbreaking entrepreneurial ventures.

At UMKC, faculty can support graduate students in entrepreneurship by providing mentorship, guiding them in applying for funding opportunities, and collaborating on business development strategies.

Additionally, UMKC's Innovation Center offers resources to help connect students with experienced entrepreneurs who can assist in launching new companies. By forming teams that combine technical expertise with business acumen, faculty members and graduate students can translate research into viable startups while ensuring that the scientific rigor of their discoveries is maintained.

This approach benefits both faculty and students. Graduate students gain valuable hands-on experience in commercialization, improving their career prospects whether they choose to stay in academia or transition to industry. Faculty members, meanwhile, can see their research make a tangible impact while maintaining their focus on advancing scientific knowledge.

Exit Strategies for Faculty Entrepreneurs

Planning an exit strategy is essential for faculty entrepreneurs who want to transition their startup to long-term sustainability. Exit strategies may include selling the company to a larger corporation, licensing the technology to an industry partner, or transitioning leadership to an external CEO while maintaining a scientific advisory role.

Faculty members should consider their long-term goals early in the commercialization process. If their priority is to remain in academia, structuring the startup in a way that allows for a transition of leadership is crucial.

The UMKC Office of Commercialization can provide guidance on structuring equity arrangements and licensing agreements to ensure that the intellectual property continues to generate value beyond the faculty member's direct involvement.

Conflict of Interest Policies for Faculty Entrepreneurs

Navigating conflicts of interest is essential for faculty members engaged in entrepreneurial activities.

The University of Missouri System's Policy on Conflict of Interest (CRR 330.015) establishes guidelines to ensure transparency, integrity, and compliance with university expectations. Faculty must proactively disclose any external interests that may intersect with their university roles to maintain ethical standards and institutional trust.

Key Considerations for Faculty Entrepreneurs

- **Disclosure Requirements:** Faculty members must disclose any financial interests, business affiliations, or external commitments related to their university responsibilities. This includes ownership in startups, consulting work, and any agreements with third parties that could present a conflict of interest.
- **Use of University Resources:** University facilities, equipment, and confidential research must not be used for personal business ventures without prior approval. Faculty should ensure that their entrepreneurial activities remain separate from their academic responsibilities.
- **Balancing Roles and Responsibilities:** Faculty entrepreneurs should establish clear distinctions between their academic and business engagements. Avoiding situations where personal financial interests could influence research, teaching, or university decision-making is critical.
- **Compliance with Federal and Institutional Regulations:** Faculty involved in federally funded research must adhere to additional conflict of interest disclosure and management requirements. CRR 410.020 and CRR 420.030 outline these expectations, ensuring compliance with national research integrity standards.

By understanding and adhering to these policies, faculty entrepreneurs can successfully navigate potential conflicts of interest while maintaining professional integrity and institutional compliance. The UMKC Innovation Center and the Office of Commercialization can provide guidance on best practices and proper disclosure procedures.

Getting Started at UMKC

If you're working through the UMKC Impact & Translation Series, these are natural next steps to begin exploring whether academic entrepreneurship is a fit for your work:

- **Leverage Institutional Resources -** UMKC provides various resources to support academic entrepreneurship, including the UMKC Innovation Center, the Office of Commercialization, the Office of Technology Transfer, CAFE, and various grant opportunities.

- **Network and Collaborate** - Engage with the entrepreneurial ecosystem both within and outside UMKC. Attend workshops, seminars, and networking events at the UMKC Innovation Center to meet like-minded individuals and potential partners.
- **Seek Mentorship** - Experienced entrepreneurs and faculty members can offer valuable insights and guidance. Mentorship programs like UMKC's Regnier Mentors connect you with seasoned professionals who can help you navigate the entrepreneurial journey.
- **Focus on the End-User** - Always keep the potential impact on the end-user in mind. This focus will help guide your research and development efforts towards creating solutions that meet real-world needs.

By engaging with these resources, researchers can bridge the gap between laboratory discoveries and market-ready innovations.

Guiding Question:

If your work were to move forward through entrepreneurship, what problem would it solve, who would it serve, and what would need to happen next?

Conclusion

This is not about becoming an entrepreneur. It is about understanding whether entrepreneurship is the right pathway for your work.

Your research already has value. The question is how that value might move beyond the university—and what it would take to get it there.

The UMKC Impact & Translation Series is designed to help you think more clearly about that path. The Office of Commercialization is here to help you explore it.

Resources & Contacts

UMKC Office of Commercialization

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